

Algorithms – Exercise Sheet 2

Discussion on Monday, 2026-05-04

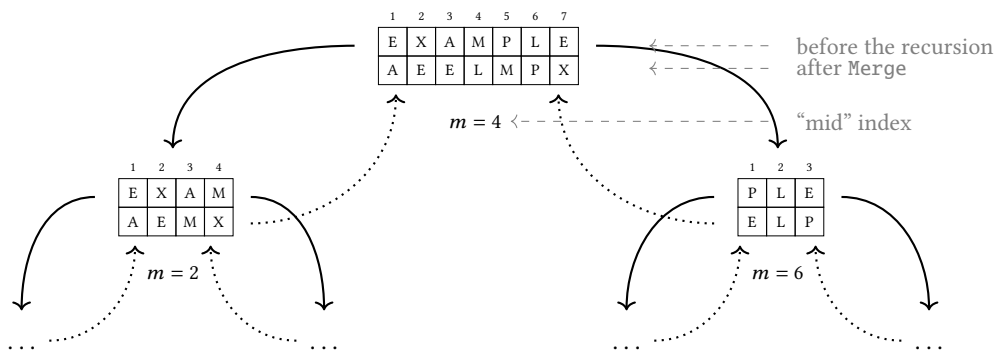
Exercise 1 (Mergesort)

Given is the following array with keys from $U = \{A, B, \dots, Z\}$:

B	U	C	K	E	T	S	O	R	T
1	2	3	4	5	6	7	8	9	10

Sort the array in lexicographically increasing order (i.e., alphabetically) with Mergesort (Slides 2.3 - 2.5) Describe the necessary intermediate steps of this algorithm (use the following figures as guide).

Note: Comments are gray. Mergesort:



Exercise 2 (Analyzing a Program)

- (a) Consider this algorithm $\mathbf{f}(A)$, which takes an array $A = A[1 \dots n]$ as input. Formulate the runtime of $\mathbf{f}$ as a recurrence depending on n !

```

Procedure  $\mathbf{f}(A[1, \dots, n])$ 
1  for  $1 \leq j \leq n$  do
2     $u = 4 \cdot j$ ;
3     $v = 5$ ;
4     $u = u + \mathbf{f}(A[1, \dots, \lfloor \frac{n}{2} \rfloor])$ ;
5    for  $1 \leq i \leq v \cdot n$  do
6       $u = u + \mathbf{f}(A[1, \dots, \lfloor \frac{n}{3} \rfloor])$ ;
7     $u = u + \mathbf{f}(A[\lfloor \frac{n}{2} \rfloor + 1, n])$ ;
8  Return  $u$ ;

```

- (b) Given the algorithm $\mathbf{g}$ with $d = 0$. Formulate the number of printed lines as a recurrence relation!
- (c) How many lines are printed by this algorithm asymptotically? To this goal, approximate the solution to the recurrence relation! You may assume that n is a power of 3, and $d = 0$.

```

Procedure  $\mathbf{g}(n, d)$ 
1   $k = 1$ ;
2   $a = 0$ ;
3  while  $k < n$  do
4     $k = k + a$ ;
5     $a = a + 1$ ;
6     $\text{print\_line}(\text{"branch"})$ ;
7  if  $k \leq 1$  then
8     $\text{print\_line}(\text{"leaf"})$ ;
9  else
10   for  $1 \leq i \leq 3^d \cdot n$  do
11      $\mathbf{g}(\lfloor \frac{n}{3} \rfloor, d + 1)$ ;

```

Exercise 3 (Karatsuba Multiplication)

Use the Karatsuba integer multiplication algorithm to multiply the two binary integers $x = 1001$ and $y = 1011$.